## Investigation: MUNDIAL S.A. - PRODUTOS DE CONSUMO (CVM code 5312), fiscal year 2011, as it was on 2024-12-31

Statistical signal based on public financial statements. Not an accusation, not investment advice.

## Company

Sector: Têxtil e Vestuário (origin: FCA); group: non-financial.
Before 2019, the company was called MUNDIAL S.A - PRODUTOS DE CONSUMO.
Trading codes (2023 FCA, version 1, filed on 2023-02-09): MNDL3 (Ações Ordinárias, Bolsa, since 1979-12-12).

## Document used

2011 DFP, version 3, filed on 2012-04-09; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2012-03-31); 2 (2012-04-02); 3 (2012-04-09).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=16056

## Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −2.915 (index)
- Alert by the study 1 rule: no
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: 28 (n = 23)
- Percentile in group and year: 18 (n = 371)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 0.957 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 38 (n = 25)
- Percentile in group and year: 34 (n = 467)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.032 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 70 (n = 25)
- Percentile in group and year: 61 (n = 477)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 0.937 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 34 (n = 25)
- Percentile in group and year: 26 (n = 472)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 0.995 (index)
- M-score component; it has no rule of its own

Percentile in sector and year: 38 (n = 25)
Percentile in group and year: 22 (n = 478)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: 1.023 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 59 (n = 23)
Percentile in group and year: 65 (n = 385)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 1.051 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 38 (n = 25)
Percentile in group and year: 58 (n = 471)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.071 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 34 (n = 25)
Percentile in group and year: 32 (n = 494)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 1.143 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 42 (n = 25)
Percentile in group and year: 73 (n = 486)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: 2.7 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 35.1 days)
Percentile in sector and year: 50 (n = 25)
Percentile in group and year: 53 (n = 449)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 32.65 p.p.
Alert by the study 1 rule: yes
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 82 (n = 25)
Percentile in group and year: 86 (n = 438)

The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.071 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 34 (n = 25)
Percentile in group and year: 34 (n = 494)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

## Financial risk (context only)

Altman Z" (financial risk, context only)
Value: −1.112 (index)
Components: X1 = −0.162; X2 = −0.064; X3 = 0.017; X4 = 0.047 (indices)
Percentile in sector and year: 34 (n = 28)
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z"
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

## Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 3/9 (8 computable)
Signals:
- Return on assets (ROA): 0
- Operating cash flow on assets (CFO): 1
- Change in ROA (ΔROA): 0
- Cash above earnings (ACCRUAL): 1
- Change in leverage (ΔLEVER): 1
- Change in current ratio (ΔLIQUID): 0
- No equity issuance (EQ_OFFER): —
- Change in gross margin (ΔMARGIN): 0
- Change in asset turnover (ΔTURN): 0
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

## Changes

Highlighted in the following year's comparative: 2 (threshold of R$8.9 million, 1% of total assets of the version used). Values in R$ million:
BPA 1 Ativo Total: 887.5 in the version used (v3, filed on 2012-04-09); 842.6 in the next DFP's comparative (v3, filed on 2013-04-09); difference of −44.9. Different value in the following year's comparative (restatement or reclassification).
BPP 2.02 Passivo Não Circulante: 480.3 in the version used (v3, filed on 2012-04-09); 435.4 in the next DFP's comparative (v3, filed on 2013-04-09); difference of −44.9. Different value in the following year's comparative (restatement or reclassification).
Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets of the version used. The other differences (reclassifications between lines and smaller

differences) stay in the full list.
Full list: 47 accounts above the R$1,000 tolerance (group a, totals and key lines: 2; group b, reclassifications: 45), in the annex at the end.

## What was known on the date

- FCA 2011, v1: 2011-02-17
- FCA 2011, v2: 2011-04-03
- DFP 2010, v1: 2011-04-04
- DFP 2010, v2: 2011-04-04
- FCA 2011, v3: 2011-05-31
- DFP 2011, v1: 2012-03-31
- DFP 2011, v2: 2012-04-02
- DFP 2011, v3: 2012-04-09
- DFP 2012, v1: 2013-03-29
- DFP 2012, v2: 2013-04-03
- DFP 2012, v3: 2013-04-09

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

## Data quality

53 applicable checks; 0 failed.

## Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
Excerpt checked literally: "Em nossa opinião, as demonstrações financeiras individuais acima referidas apresentam adequadamente, em todos os aspectos relevantes, a posição patrimonial e financeira da MUNDIAL S.A. - PRODUTOS DE CONSUMO em 31 de dezembro de 2011, o desempenho de suas operações e os seus fluxos de caixa para o exercício findo naquela data, de acordo com as práticas contábeis adotadas no Brasil."
Auditor in the 2021 FCA: Baker Tilly Brasil RS Auditores Independentes Sociedade Simples.
Auditor in the 2022 FCA: Baker Tilly Brasil RS Auditores Independentes Sociedade Simples, Taticca Auditores Independentes S.S.
Auditor in the 2023 FCA: Baker Tilly Brasil RS Auditores Independentes Sociedade Simples, Taticca Auditores Independentes S.S.
Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2012: Grant Thornton Auditores Independentes -> KPMG Auditores Independentes (transition: two firms in the 2012 FCA). The outgoing firm audited 3 consecutive fiscal years (2009 to 2011), by the start and end dates in the FCA.
- 2017: KPMG Auditores Independentes -> Baker Tilly Brasil RS Auditores Independentes Sociedade Simples (transition: two firms in the 2017 FCA). The outgoing firm audited 5 consecutive fiscal years (2012 to 2016), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).
- 2022: entry of Taticca Auditores Independentes S.S, with no other firm leaving.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

## How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

## Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

## Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 38 DFP lines, each with document, version, filing date, account and location.

## Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-11T06:39:52+00:00.

## Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: MUNDIAL S.A - PRODUTOS DE CONSUMO.

## Annex: full list of differences in the following year's comparative (R$ million)

(a) BPA 1 Ativo Total: 887.5 in the version used, 842.6 in the comparative (difference of −44.9)
(a) BPP 2.02 Passivo Não Circulante: 480.3 in the version used, 435.4 in the comparative (difference of −44.9)
(b) DFC (indirect method) 6.03.01 Emprestimos e Financiamentos: −0.3 in the version used, 351.5 in the comparative (difference of 351.8)
(b) DFC (indirect method) 6.03.02 Debentures a Pagar: −7.1 in the version used, −312.3 in the comparative (difference of −305.2)
(b) BPA 1.02.01.09 Outros Ativos Não Circulantes: 0.0 in the version used, 111.1 in the comparative (difference of 111.1)
(b) BPA 1.02.01.03 Contas a Receber: 75.9 in the version used, 0.0 in the comparative (difference of −75.9)
(b) BPA 1.02.01.03.02 Outras Contas a Receber: 75.9 in the version used, 0.0 in the comparative (difference of −75.9)
(b) BPP 2 Passivo Total: 887.5 in the version used, 842.6 in the comparative (difference of −44.9)
(b) DFC (indirect method) 6.01.02.09 Impostos e Contribuições: 30.6 in the version used, −9.0 in the comparative (difference of −39.6)
(b) DFC (indirect method) 6.03.03 Participação de Acionistas Não Controladores: 0.0 in the version used, −39.5 in the comparative (difference of −39.5)
(b) BPA 1.01 Ativo Circulante: 223.4 in the version used, 187.9 in the comparative (difference of −35.5)
(b) BPA 1.01.08 Outros Ativos Circulantes: 67.4 in the version used, 31.8 in the comparative (difference of −35.5)
(b) BPA 1.01.08.03 Outros: 67.4 in the version used, 31.8 in the comparative (difference of −35.5)
(b) BPA 1.01.08.03.01 Títulos a Receber: 51.8 in the version used, 16.2 in the comparative

(difference of −35.5)
(b) BPP 2.02.02 Outras Obrigações: 393.4 in the version used, 357.9 in the comparative (difference of −35.5)
(b) BPP 2.02.02.02 Outros: 393.2 in the version used, 357.7 in the comparative (difference of −35.5)
(b) BPP 2.02.02.02.06 Obrigações Fiscais Federais: 339.4 in the version used, 303.9 in the comparative (difference of −35.5)
(b) DFC (indirect method) 6.01.01.03 Provisão para Créditos de Liquidação Duvidosas: 0.4 in the version used, 34.9 in the comparative (difference of 34.5)
(b) BPA 1.02.02 Investimentos: 27.4 in the version used, 0.5 in the comparative (difference of −27.0)
(b) BPA 1.02.02.02 Propriedades para Investimento: 27.0 in the version used, 0.0 in the comparative (difference of −27.0)
(b) DFC (indirect method) 6.01.02.04 Estoque: −1.4 in the version used, 21.6 in the comparative (difference of 23.0)
(b) DFC (indirect method) 6.01.02.06 Outras Contas a Receber - Circulante e Não Circulante: 21.6 in the version used, 1.3 in the comparative (difference of −20.3)
(b) DFC (indirect method) 6.01.02.03 Titulos a Receber: −21.3 in the version used, −1.4 in the comparative (difference of 19.8)
(b) BPA 1.02.01 Ativo Realizável a Longo Prazo: 404.4 in the version used, 422.1 in the comparative (difference of 17.6)
(b) BPA 1.02.01.06 Tributos Diferidos: 18.3 in the version used, 0.8 in the comparative (difference of −17.6)
(b) DFC (indirect method) 6.02.01 Investimentos: 0.0 in the version used, −16.2 in the comparative (difference of −16.2)
(b) DFC (indirect method) 6.02.02 Imobilizado: −16.2 in the version used, −0.2 in the comparative (difference of 16.1)
(b) DFC (indirect method) 6.01.01.04 Juros a Receber sobre Vendas de Ativos: −12.0 in the version used, 1.8 in the comparative (difference of 13.8)
(b) DFC (indirect method) 6.01.01.02 Variação Cambial Clientes: 0.9 in the version used, 13.6 in the comparative (difference of 12.7)
(b) DFC (indirect method) 6.01.01 Caixa Gerado nas Operações: −3.5 in the version used, 8.5 in the comparative (difference of 12.0)
(b) DFC (indirect method) 6.01.02 Variações nos Ativos e Passivos: 26.1 in the version used, 14.1 in the comparative (difference of −12.0)
(b) BPA 1.02 Ativo Não Circulante: 664.1 in the version used, 654.7 in the comparative (difference of −9.4)
(b) BPA 1.02.01.06.01 Imposto de Renda e Contribuição Social Diferidos: 10.1 in the version used, 0.8 in the comparative (difference of −9.4)
(b) BPP 2.02.03 Tributos Diferidos: 58.8 in the version used, 49.5 in the comparative (difference of −9.4)
(b) BPP 2.02.03.01 Imposto de Renda e Contribuição Social Diferidos: 58.8 in the version used, 49.5 in the comparative (difference of −9.4)
(b) BPA 1.02.01.06.02 Impostos a Recuperar: 8.2 in the version used, 0.0 in the comparative (difference of −8.2)
(b) DFC (indirect method) 6.01.02.08 Fornecedores: 1.3 in the version used, −4.9 in the comparative (difference of −6.2)
(b) DFC (indirect method) 6.01.02.10 Salários e Ordenados: 1.1 in the version used, −2.5 in the comparative (difference of −3.7)
(b) DFC (indirect method) 6.01.02.02 Provisão para Créditos de Liquidação Duvidosa: 0.0 in the version used, 2.3 in the comparative (difference of 2.3)
(b) BPP 2.03.04.01 Reserva Legal: 0.0 in the version used, 1.7 in the comparative (difference of 1.7)
(b) BPP 2.03.04.10 Reserva Orçamento de Capital: 23.7 in the version used, 22.0 in the comparative (difference of −1.7)
(b) DFC (indirect method) 6.01.01.05 Ajuste por Obsolecencia: 0.0 in the version used, −1.6 in

the comparative (difference of −1.6)
(b) BPP 2.03.07 Ajustes Acumulados de Conversão: 0.0 in the version used, −1.1 in the comparative (difference of −1.1)
(b) BPP 2.03.06 Ajustes de Avaliação Patrimonial: −1.0 in the version used, 0.0 in the comparative (difference of 1.0)
(b) DFC (indirect method) 6.01.02.05 Partes Relacionadas: 1.2 in the version used, 1.5 in the comparative (difference of 0.2)
(b) DFC (indirect method) 6.02.03 Intangíveis: −0.2 in the version used, 0.0 in the comparative (difference of 0.2)
(b) BPP 2.03.05 Lucros/Prejuízos Acumulados: −83.7 in the version used, −83.5 in the comparative (difference of 0.1)